



International University of Sarajevo

Faculty of Engineering and Natural Sciences

SE407 – Software Quality Management

A Governance-Oriented Framework for Software Quality Assurance in Digitally Transforming Enterprises

Group Research Report

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Academic Year 2025-2026

Abstract

Digital transformation is changing the way organizations develop and manage software, making traditional software quality assurance approaches less effective on their own. This report presents a governance-oriented framework for software quality assurance (GoF-SQA) designed for enterprises moving toward cloud-native and automated development environments. The framework focuses on identity and secrets management as a practical starting point for discussing governance improvements within software delivery pipelines.

Instead of treating governance as a separate compliance activity, the proposed framework integrates it directly into the development process through policy-as-code, role-based access control (RBAC), and automated audit evidence collection. By combining ideas from DevSecOps, software governance, and quality management research, the framework introduces a layered architecture that connects governance controls with measurable quality outcomes.

The report also defines several key performance indicators, including policy violation rates, time required to collect audit evidence, and unauthorized access attempts, allowing organizations to continuously monitor and improve their quality posture. In addition, a phased implementation roadmap is provided to help organizations adopt the framework gradually based on their level of maturity.

Through the analysis of cloud-native scenarios, the report suggests that embedding governance directly into the software delivery pipeline may improve software reliability and audit readiness. At the same time, it recognizes challenges such as organizational readiness and differences between existing toolchains. Overall, the framework demonstrates how governance can become a practical and measurable part of modern software quality assurance rather than just an external compliance requirement.

1. Introduction

Digital transformation has become one of the main priorities for modern enterprises. As organizations migrate workloads to cloud-native architectures, adopt microservice designs, and accelerate release cycles through continuous delivery pipelines, the underlying assumptions governing software quality assurance must be re-examined. In this environment, quality can no longer be checked only at fixed stages of development; it must be embedded as a continuous, automated concern throughout the entire software development life cycle (SDLC).

Traditional quality management standards such as ISO 9001 and CMMI provide valuable structural guidance but were largely conceived in a context of waterfall development and on-premises infrastructure. In digitally transforming enterprises, the deployment frequency, and distributed team structures that characterize modern engineering create governance gaps that these frameworks do not fully address. Specifically, the question of who has access to what system, under what conditions, and with what level of auditability, has become a central quality concern rather than a peripheral security consideration.

Identity and secrets management represents a particularly instructive lens through which to examine governance-oriented quality assurance. In a continuous delivery environment, build pipelines, deployment agents, API integrations, and runtime services all require credentials, certificates, tokens, and cryptographic keys. The growing number of these secrets, combined with the fast pace of cloud-native environments, creates measurable quality risks: unauthorized deployments, unrotated credentials exploited by adversaries, unsigned artifacts introduced into production, and audit logs insufficient to support compliance reviews.

This report advances the argument that a governance-oriented framework, when designed with identity and secrets management as its foundational layer, provides a practical and measurable approach to software quality assurance in digitally transforming contexts. Rather than treating governance as a post-development compliance exercise, the proposed GoF-SQA framework embeds governance controls as first-class engineering artifacts, enabling organizations to measure, automate, and continuously improve their quality posture. The remainder of this report is organized as follows: Section 2 reviews related work in software governance and quality frameworks; Section 3 presents the methodology and framework design; Section 4 defines quality metrics and the evaluation model; Section 5 outlines the implementation roadmap; Section 6 provides discussion and analysis; and Section 7 concludes the report.

2. Related Work

2.1 Software Quality Models in the DevOps Era

The intersection of DevOps practices with traditional software quality models has attracted significant scholarly attention. Lwakatare et al. (2016) examined the relationship between DevOps, agile, lean, and continuous deployment, showing that DevOps extends earlier software delivery paradigms by emphasizing collaboration, automation, and continuous delivery practices. More recently, Forsgren et al. (2018), in *Accelerate*, demonstrated that high-performing software delivery organizations are distinguished not merely by speed, but also by their ability to maintain stability, reliability, and rapid recovery. These findings suggest that modern software quality depends not only on testing activities, but also on governance mechanisms that support reliable and controlled delivery.

The McCall Quality Model, originally formulated in the late 1970s, and subsequent models, including Boehm's quality taxonomy, were foundational in establishing structured dimensions of software quality. However, classical quality models require reinterpretation in cloud-native contexts, particularly where observability, access accountability, dynamic infrastructure, and automated governance become central to quality assurance. The ISO/IEC 25010:2023 product quality model provides a more contemporary footing by defining a product quality model for ICT and software products, organized around quality characteristics and sub-characteristics. Nevertheless, ISO/IEC 25010 does not, by itself, prescribe operational mechanisms for governing dynamic access rights, secrets, or ephemeral infrastructure in containerized environments.

2.2 Governance Frameworks in Software Engineering

The IT governance literature, particularly COBIT 2019 (ISACA, 2019) and ISO/IEC 38500:2024, establishes principles for aligning technology initiatives with organizational objectives through defined accountability structures. ISO/IEC 38500:2024 provides guiding principles for governing bodies and those who support them in ensuring the effective, efficient, and acceptable use of information technology within organizations. Kim et al. (2016), in *The DevOps Handbook*, articulate the three principles of flow, feedback, and continual learning, which are aligned with governance goals but do not provide the granular technical mechanisms needed for quality assurance in secrets-intensive environments.

Humble and Farley (2010) introduced the deployment pipeline as a mechanism for making build, test, release, and deployment activities repeatable and reliable. While this work predates the wide adoption of policy-as-code tooling, it established an important foundation for treating quality and compliance checks as automated pipeline stages. Subsequent work by Bass et al. (2015) extended this thinking to architectural governance by emphasizing that quality attributes must be explicitly designed into system structures rather than tested only after implementation.

2.3 Identity and Secrets Management as a Quality Concern

The management of machine identities and cryptographic secrets in software systems has evolved from a peripheral operational concern to a recognized quality and security governance imperative. GitGuardian's *State of Secrets Sprawl 2024* report found that 12.8 million new secret occurrences were detected in public GitHub commits in 2023, representing a 28% increase from 2022. This illustrates the continuing risk of hardcoded credentials and unmanaged secrets in modern software delivery. Arundel and Domingus (2019), in *Cloud Native DevOps with Kubernetes*, treat secrets management as a foundational concern for production-grade systems, advocating for external secret stores and least-privilege access as engineering baselines.

From a standards perspective, NIST Special Publication 800-204C (Chandramouli, 2022) provides guidance on implementing a DevSecOps strategy for cloud-native applications. The CIS Kubernetes Benchmark similarly provides secure configuration guidance for Kubernetes environments, including controls relevant to cluster hardening and workload security. Despite this attention, the integration of secrets governance into broader quality assurance frameworks as measurable, sprint-level quality indicators remains underdeveloped in academic software quality literature.

2.4 Policy-as-Code and Automated Compliance

2.4 Policy-as-Code and Automated Compliance

Policy-as-code, the practice of expressing compliance and governance rules as machine-executable artifacts, has gained substantial traction with tools such as Open Policy Agent (OPA) and HashiCorp Sentinel. In software delivery environments, policy-as-code allows governance requirements to be expressed as version-controlled rules that can be evaluated automatically during code review, infrastructure provisioning, deployment admission, and runtime compliance checks.

This approach is important for software quality assurance because it shifts governance from periodic manual review toward continuous automated enforcement. Rather than treating compliance as a separate audit activity, policy-as-code allows governance checks to become part of the same CI/CD workflow used for testing, building, and deploying software. These capabilities help motivate the policy enforcement layer in the proposed GoF-SQA framework.

Taken together, the reviewed literature suggests that modern SQA requires more than defect detection and process compliance. In cloud-native environments, quality assurance must also account for identity governance, secrets lifecycle management, automated policy enforcement, and audit-oriented observability. These four concerns form the conceptual basis for the GoF-SQA framework presented in the next section.

3. Methodology and Quality Framework Design

3.1 Research Methodology

This report employs a design science research (DSR) methodology, following the guidelines articulated by Hevner et al. (2004). DSR is appropriate for this context because the research objective is to construct an artifact, in this case a governance framework, that solves an identified problem in a specific class of organizations. The methodology proceeds through problem identification, objective definition, framework design, scenario-based analysis, discussion against defined criteria, and communication of findings. Secondary literature synthesis and qualitative scenario modeling were the primary methods employed.

The scope of the framework was bounded to digitally transforming enterprises, defined as organizations in active transition from traditional, monolithic software delivery to cloud-native, automated delivery models. This scope was chosen because these organizations often experience the most governance gaps: organizations in this state simultaneously carry legacy compliance obligations and newly introduced DevOps complexities, making governance coherence particularly challenging and valuable.

3.2 Framework Architecture Overview

The proposed Governance-Oriented Framework for Software Quality Assurance (GoF-SQA) is conceptually organized as a four-layer architecture. Each layer addresses a distinct governance concern while contributing to the overall quality assurance posture. The layers are: (1) Identity and Access Governance, (2) Secrets Lifecycle Management, (3) Policy Enforcement and Automation, and (4) Audit Evidence and Observability.

This layered design was intended to allow independent adoption. Organizations at lower maturity levels can implement the foundational Identity and Access Governance layer without immediately requiring the full automation infrastructure of the upper layers. The framework is therefore not prescriptive about implementation order but is prescriptive about the completeness criteria at each layer before advancement.

3.3 Layer 1: Identity and Access Governance

The first layer establishes the foundational principle that every actor in the software delivery system, including human engineers, service accounts, build agents, and runtime workloads, must possess a verified, minimally privileged identity. Role-based access control (RBAC) definitions are treated as version-controlled engineering artifacts, subject to code review, approval workflows, and automated drift detection. Privileged identities, defined as those capable of modifying production infrastructure or approving deployments, are subject to mandatory multi-factor authentication and time-bounded access grants.

A critical quality control in this layer is the prohibition of shared credentials and the requirement that build agents operate under ephemeral, scoped identities rather than long-lived service account tokens. This control directly addresses a common failure mode in digitally transforming enterprises where CI/CD systems inherit the permissions of their human administrators, creating excessive blast radius in the event of credential compromise.

3.4 Layer 2: Secrets Lifecycle Management

The second layer governs the creation, distribution, rotation, and revocation of all cryptographic secrets used in the software delivery pipeline. The framework mandates the use of centralized secrets management infrastructure such as HashiCorp Vault, AWS Secrets Manager, or Azure Key Vault, with dynamic secret generation preferred over static, pre-provisioned credentials wherever the target system supports it. All secrets must carry metadata, including owner, expiration date, last rotation timestamp, and consuming service identifier.

A mandatory quality gate in this layer is the secrets rotation policy: no secret used in a production deployment may exceed a configurable maximum age, typically defined as 90 days for long-lived credentials and 24 hours for dynamically generated tokens. In practice, violations of this policy could trigger automated pipeline blocks and be recorded as quality metric events. The framework further proposes that production deployments should be signed with cryptographic attestations, ensuring supply chain integrity from code commit to running artifact.

3.5 Layer 3: Policy Enforcement and Automation

The third layer implements governance intent through machine-executable policy artifacts. Using policy-as-code frameworks, the governance team, in collaboration with security and engineering stakeholders, defines mandatory controls such as prohibited deployment patterns, required labels on infrastructure resources, and mandatory approval chains for sensitive changes. These policies can be enforced at multiple points in the delivery pipeline, including code review, artifact build time, deployment admission, and runtime compliance scanning.

The policy-as-code approach yields two quality benefits beyond enforcement: it makes governance intent transparent and auditable, and it creates a natural mechanism for detecting policy drift between the stated governance posture and actual system configurations. Regular automated reconciliation processes can compare declared policy states against observed production configurations, helping organizations generate compliance reports with minimal manual effort.

3.6 Layer 4: Audit Evidence and Observability

The fourth layer ensures that all governance control activities produce structured, queryable audit evidence. In a practical implementation, policy evaluations, access control decisions, secret rotation events, and deployment attestations would generate tamper-evident log entries in a centralized logging system. These entries can be formatted using structured schemas aligned

with common audit frameworks such as SOC 2 Type II and ISO/IEC 27001 to support automated evidence collection.

Observability in this context extends beyond application performance monitoring to encompass governance observability: dashboards that surface policy violation rates, access anomalies, secrets age distributions, and deployment signing compliance in real time. This visibility could support both engineering teams seeking fast feedback and audit teams requiring evidence packages, creating a shared artifact that serves dual purposes without duplicating effort.

4. Quality Metrics and Evaluation Model

4.1 Metric Design Principles

The evaluation model for GoF-SQA is grounded in the Goal-Question-Metric (GQM) paradigm introduced by Basili and Weiss (1984). GQM operates on three hierarchical levels: at the conceptual level, a governance goal is articulated; at the operational level, questions are derived that characterize the goal; and at the quantitative level, metrics are defined. Four design principles govern metric selection: Automated collectability, Actionability, Layer traceability, and Non-redundancy.

4.2 Core Governance Quality Indicators

The framework defines six primary quality indicators, each mapped to one of the four framework layers. These indicators use a two-tier threshold alerting system (Warning and Critical) to ensure immediate response to quality deviations.

Code	Metric Name	Target	Weight	Warning Threshold	Critical Threshold	Layer
PVRS	Policy Violation Rate per Sprint	< 2 per 100 executions	20%	2–5 violations	> 5 violations	Layer 3
TTAE	Time-to-Audit-Evidence	< 4 hours	15%	4–12 hours	> 12 hours	Layer 4
UAAR	Unauthorized Access Attempt Rate	Downward trend	15%	Sudden increase	Persistent escalation	Layer 1
SACR	Secrets Age Compliance Rate	$\geq 95\%$	25%	90–95%	< 90%	Layer 2
DSC	Deployment Signing Coverage	100%	15%	95–99%	< 95%	Layer 2
IDS	Identity Drift Score	0 discrepancies	10%	1–3 discrepancies	> 3 discrepancies	Layer 1

4.3 Evaluation Model

Governance Quality Index (GQI) Formula

The Governance Quality Index (GQI) is defined as a weighted composite score derived from six governance quality indicators. Each metric is normalized to a value between 0 and 100 and multiplied by its assigned governance weight.

$$GQI = (PVRSn \times 0.20) + (TTAEn \times 0.15) + (UAARn \times 0.15) + (SACRn \times 0.25) + (DSCn \times 0.15) + (IDSn \times 0.10)$$

Example Calculation

Assume the following monthly normalized scores:

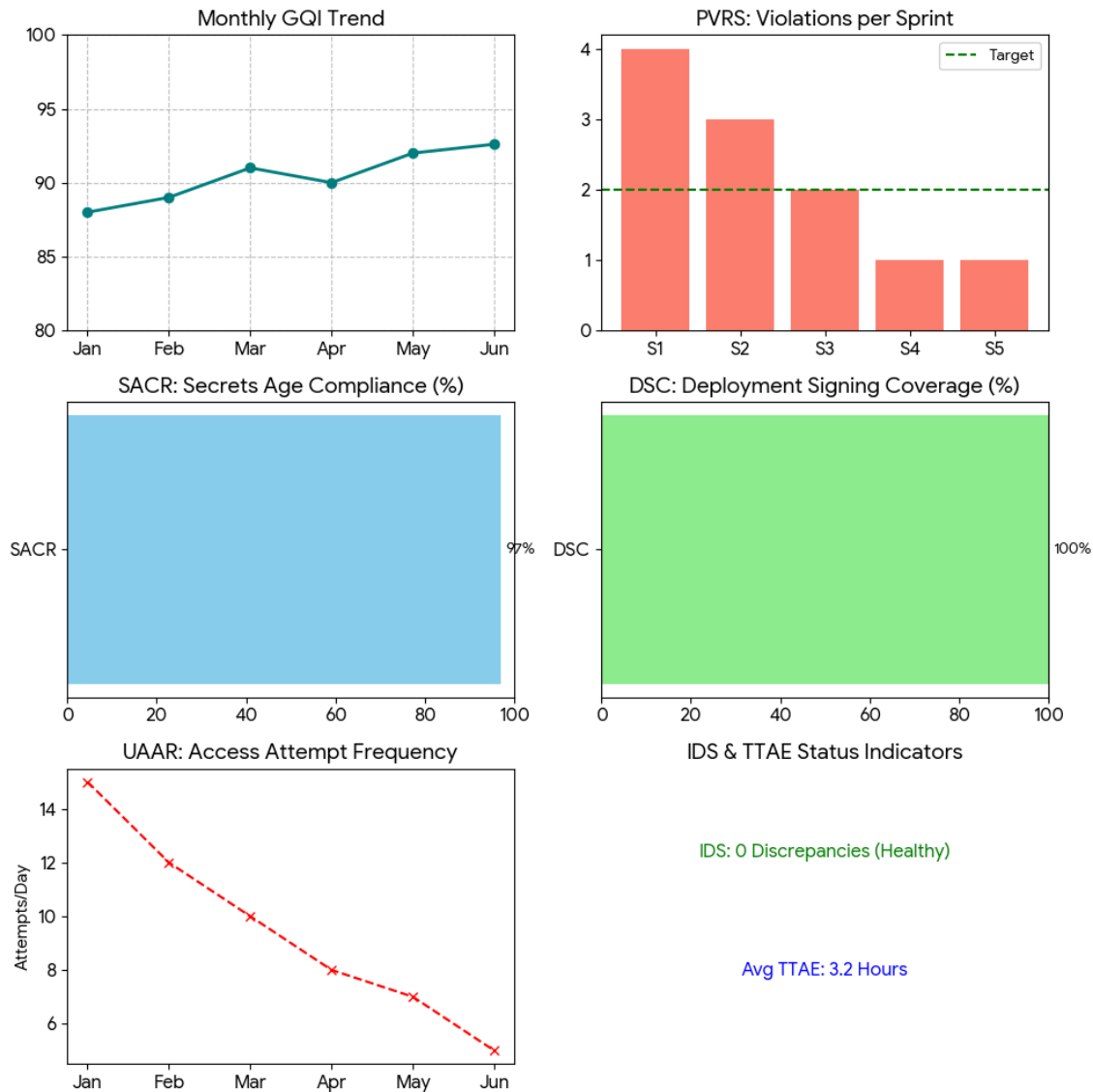
Metric	Score	Weight
PVRS	92	20%
TTAE	85	15%
UAAR	88	15%
SACR	97	25%
DSC	100	15%
IDS	90	10%

$$GQI = (92 \times 0.20) + (85 \times 0.15) + (88 \times 0.15) + (97 \times 0.25) + (100 \times 0.15) + (90 \times 0.10)$$

$$GQI = 18.4 + 12.75 + 13.2 + 24.25 + 15 + 9 = \mathbf{92.6}$$

Within the proposed model, a GQI above 90 would represent a highly mature governance-oriented quality assurance posture with strong compliance and audit readiness characteristics.

Governance Quality Dashboard



Example governance observability dashboard illustrating possible GQI trends and metric distributions.

5. Development and Implementation Roadmap

5.1 Phase 1: Foundation (Months 1 to 3)

The first phase of implementation focuses on preparing the organization for the framework. Before introducing advanced automation, the organization first needs to understand its current governance situation. This includes checking how access rights are managed, where secrets are stored, how service accounts are used, and whether the CI/CD pipeline already has some basic security and quality controls.

In this phase, a governance maturity assessment should be performed using questions related to the four layers of the framework. The organization should also document existing secrets, credentials, service accounts, and permissions used in the software delivery process. This is important because many governance problems happen when old credentials or unnecessary access rights remain active without being reviewed.

Another important activity is forming a small cross-functional Governance Working Group. This group should include people from engineering, security, compliance, and operations. Governance should not be handled by only one team, because software quality depends on cooperation between all teams involved in development and deployment.

During this phase, the organization should also choose suitable tools for identity and secrets management. For organizations mainly using one cloud provider, native tools such as AWS IAM with AWS Secrets Manager or Azure Active Directory with Azure Key Vault can be practical choices. For organizations using multiple cloud providers or hybrid infrastructure, HashiCorp Vault may be more suitable because it is less tied to one vendor.

5.2 Phase 2: Automation Integration (Months 4 to 6)

The second phase focuses on connecting the governance controls with the CI/CD pipeline. After the basic identity and secrets management controls are prepared, the organization can start automating them.

In this phase, policy-as-code rules can be introduced. These rules may check whether secrets are rotated on time, whether RBAC permissions are too broad, or whether deployment artifacts are properly signed. Instead of checking these things manually, the pipeline can automatically detect problems and block risky changes before they reach production.

Audit logging should also be improved during this phase. Events such as policy violations, secret rotations, deployment approvals, and access changes should be collected in a centralized logging or monitoring system. This makes it easier to prepare audit evidence when needed and also helps teams understand where governance problems are happening.

At the end of this phase, teams should have access to dashboards that show the main governance quality metrics. These dashboards can help developers and managers follow indicators such as policy violation rate, secrets compliance, unauthorized access attempts, and deployment signing coverage.

5.3 Phase 3: Continuous Improvement (Months 7 to 12)

The third phase focuses on improving the framework over time. Governance should not be treated as a one-time project that is finished after the tools are installed. It should become part of the normal software development process.

During this phase, the Governance Quality Index and individual metrics should be reviewed regularly. If some metrics show negative trends, the organization should investigate the cause and define corrective actions. For example, if the secrets age compliance rate becomes lower, the team should check whether some applications are still using old credentials or whether the rotation process is not automated enough.

The policy library can also be expanded in this phase. After the basic policies are stable, the organization can add new rules for other governance areas, such as infrastructure configuration, production approvals, or access reviews.

More advanced practices can also be introduced, such as automated identity lifecycle management. This means that user and service account access can be automatically created, updated, or removed based on organizational changes. For example, when an employee changes role or leaves the organization, their access can be updated automatically instead of relying only on manual work.

Organizations may also perform controlled red-team exercises to test how well the framework detects unauthorized access attempts, identity drift, and other governance weaknesses.

To make the implementation process more practical, the proposed GoF-SQA framework is divided into three phases. Each phase focuses on a different level of adoption, starting with basic governance foundations and gradually moving toward automation and continuous improvement.

Phase	Time Period	Main Focus	Key Activities
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Phase 1: Foundation	Months 1–3	Preparing the organization for governance adoption	Governance maturity assessment, RBAC setup, secrets inventory, formation of Governance Working Group
Phase 2: Automation Integration	Months 4–6	Connecting governance controls with the CI/CD pipeline	Policy-as-code rules, secrets rotation checks, RBAC validation, automated audit logging
Phase 3: Continuous Improvement	Months 7–12	Monitoring and improving governance quality over time	GQI review, dashboards, root-cause analysis, policy library expansion, red-team exercises

6. Discussion and Analysis

6.1 Strengths of the Proposed Framework

One of the main strengths of GoF-SQA is that it connects governance with real engineering practices. Instead of keeping governance only in documents, checklists, or audit reports, the framework tries to include it directly in the development and deployment process.

This is useful because modern software teams already use version control, CI/CD pipelines, automated testing, and monitoring tools. By treating RBAC rules, secret rotation policies, and policy-as-code rules as engineering artifacts, governance becomes easier to manage and review. Developers do not need to work in a completely separate compliance system, because governance controls can become part of the same workflow they already use.

Another strength is that the framework is measurable. Many governance problems are difficult to discuss because they are only described in general terms. In this framework, indicators such as policy violation rate, secrets age compliance, identity drift, and time-to-audit-evidence make governance more visible. This allows teams to track whether their quality posture is improving or getting worse over time.

The phased roadmap is also a practical advantage. Organizations do not need to implement the whole framework at once. They can start with basic identity and secrets management, then later introduce automation, dashboards, and continuous improvement activities. This makes the framework more realistic for enterprises that are still in the middle of digital transformation.

6.2 Limitations and Constraints

Even though the framework has several advantages, it also has some limitations. First, it assumes that the organization already has a certain level of DevOps maturity. For example, automated CI/CD pipelines, version-controlled infrastructure, and centralized logging are almost necessary for the framework to work properly. If an organization still depends mostly on manual deployments, it may be difficult to apply the automation layers immediately.

Second, the framework focuses more on technical governance controls than on organizational culture. Tools and policies can help, but they are not enough if teams see governance only as extra bureaucracy. For the framework to work, development teams need to understand that policy violations are quality signals, similar to failed tests or security warnings. This requires support from management and good communication between engineering, security, and compliance teams.

Third, the Governance Quality Index can be useful as a summary score, but it should not be the only thing used for decision-making. A high GQI may hide problems in one specific area. For example, an organization may have good deployment signing coverage and low policy violation

rates, but still have weak secrets rotation. Because of this, individual metrics should always be reviewed together with the final GQI score.

Another limitation is that different organizations have different tools and processes. A framework that works well in one company may require changes before it can be applied in another company. For example, a company using AWS, GitHub Actions, and Kubernetes will implement the framework differently from a company using Azure DevOps and virtual machines.

6.3 Applicability Across Industries

Although GoF-SQA is mainly designed for digitally transforming enterprises, it can be useful in several industries. Financial services, healthcare, public administration, and technology companies all depend on software systems where access control, secrets management, audit evidence, and reliability are important.

In regulated industries, the framework may be especially useful because organizations need to prove that their systems are controlled and auditable. For example, banks need strong access control and evidence for compliance, while healthcare organizations must protect sensitive patient data. Public institutions also need transparency and accountability when managing digital services.

The layered structure makes the framework flexible. Organizations can adopt only the layers that are most relevant at the beginning and expand later. Also, specific regulations or internal policies can be added as policy-as-code rules without changing the entire framework.

Overall, the framework is not a complete solution for every organization, but it provides a structured way to connect governance, software quality, and automation in modern software delivery environments.

7. Conclusion

This report argues that software quality assurance in digitally transforming enterprises requires a governance orientation that treats identity management, secrets governance, and automated policy enforcement as first-class quality engineering concerns. The proposed GoF-SQA framework provides a structured, four-layer architecture that operationalizes this orientation through concrete technical controls, measurable quality indicators, and a phased implementation roadmap.

The framework's central argument is that governance does not need to be treated as an external constraint imposed on software engineering teams, but can instead be an integrated, automated dimension of the delivery pipeline that produces quality improvements alongside compliance artifacts. Aligning governance activities with sprint-level metrics, such as policy violation rates, secrets compliance, and deployment signing coverage, may help engineering teams respond more effectively to governance-related quality signals with the same immediacy that they apply to code quality and test coverage signals.

Future research directions include empirical validation of the GoF-SQA framework through longitudinal case studies and real-world organizational analysis, investigation of the interaction effects between organizational maturity and framework adoption outcomes, and extension of the policy-as-code component to encompass AI-assisted anomaly detection for governance observability. As enterprises continue their digital transformation journeys, the convergence of software quality assurance and technology governance will likely deepen, and frameworks capable of addressing both dimensions in an integrated manner will become increasingly essential. Taken together, these elements suggest that governance-oriented quality assurance may improve reliability, audit readiness, and delivery consistency over time.

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